How Test-Set Peeking Inflates Benchmark Comparisons

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***Abstract*—People often compare a new method to "the best single model," picked only after looking at each test's score. That is not fair — it is like picking your answer after seeing the answer key. This best model wins partly by luck, not by truly being better, which makes the new method look worse than it really is. I study that unfair comparison here, not the new method being tested.**

I built a machine-learning pipeline called CRAFT and ran it on seven datasets, each split 30 different ways. Against it, I compared two ways of picking the best model. One way, the naive way, is the oracle: it picks after looking at the test score. The other way, the stable way, never looks at the test score; it picks using a different check called cross-validation. On five datasets, both ways agree exactly, with no unfair advantage at all, just as a simple math formula predicts. On the other two datasets, AI Jobs and HR Attrition, the naive way's unfair advantage is big enough to change the result of a fairness test. Once I correct for running many such tests at once, though, only HR Attrition's result still holds up, and even that one rests on an assumption I show openly instead of hiding. This unfair advantage is not just a math trick. On three large datasets, measured directly against about 40,000 extra rows, the naive method's reported score was 0.0041–0.0191 higher than its real score, and the stable method had the same problem, just smaller.

Splits from the same dataset are not fully separate from each other, so I checked the same idea somewhere else too: 163 different public datasets, from three big benchmark collections (TabRepo, TabArena, TabZilla). There, just swapping which best model you compare against changes which model comes out ahead in 31% of cases. The problem grows in exactly the way I expected: worse with more choices to pick from, worse when the choices are close to each other, and worse when the check used to pick is weaker.

A best model picked after seeing the test score is not a fair, neutral comparison. It should be treated as a ceiling — the best possible score, not a realistic one. Papers should report how big that extra advantage likely is, not just whether the difference looks "significant."

***Index Terms*—benchmark selection bias, oracle baseline, model comparison, repeated random splits, Nadeau–Bengio correction, placebo test, tabular benchmarks**

I. Introduction

Papers often compare a new method with "the best single model." This sounds fair, but it can hide a problem. Imagine you pick the best model only after you see how each test turned out: that pick then gets some credit just from luck. This best model ends up looking stronger than it really is, and the new method looks weaker by comparison. A pick made this way is an oracle, not a real competitor — like someone who already saw the answer key. On every test, it matches the best of its choices; no real method, one that must choose before seeing the answer, can do that.

Scientists have known about this problem for a long time. If you pick the best of several noisy scores, the winner's score ends up too high [1], [2]. This happens when picking a model [3], and when measuring error rates too [4], [5]. An unfair test setup can even flip the answer to whether something is really better [3]. Recent studies found the same problem at a bigger scale: one pushed up an AutoML system's win count across 513 datasets [6], another found the same trick beats a random pick across thousands of datasets [7], and sometimes, which model family looks best flips completely once you check it a different way [8]. I do not claim to be the first to find this problem.

What people have studied less is what happens when this peeking happens on the comparison side, not the new method's side. That raises three questions: how much does peeking puff up the score? When does that change whether the new method looks better or worse? And which assumption about the data decides the answer? I study these three questions using CRAFT, a machine-learning pipeline I built for a larger research project, after checking my own pipeline and finding a problem: its best-single-model comparison was picked after peeking at that round's test score, every single time.

My test changes only one thing. CRAFT's predictions stay the same. The way the data gets split into training and testing also stays the same. Only one rule changes: how the best model gets picked. One way (naive) picks after seeing the test score. The other way (stable) picks using cross-validation, without ever seeing the test score. Because only the rule for picking the best model changes, every comparison is fair and matched.

I make five contributions.

1. **How big the unfair advantage is, not just whether it is "significant."** For each dataset, I report the gap between the naive and stable picks, plus a range showing how sure I am. I also use a 65-year-old math formula (Clark's) that predicts how much luck the oracle should get — and I check it against data I locked in before running anything.
2. **Where the problem shows up.** How close two candidate models are, measured against how noisy their scores are, predicts which datasets will show this problem. Five of my seven datasets act like a control group: the problem correctly disappears on them.
3. **The one assumption that decides the answer.** Two standard statistical tests disagree with each other here. Instead of just picking one, I show how the answer changes depending on one assumption — how related

the repeated tests are to each other — and mark the exact point where the answer flips.

4. **Tests where I already know the right answer.** I ran four checks where the correct result was already known ahead of time: a placebo test, a test with the labels shuffled randomly, a test against real ground truth on a huge extra pool of data, and a computer simulation. All four confirm the effect and the tests are behaving correctly.
5. **A test using many different datasets.** Different datasets, unlike repeated splits of the same one, are truly separate from each other. Using public results from 163 datasets, I measure how the unfair advantage grows: with more candidates to choose from, with candidates that are closer together, and with a weaker way of picking. I also measure how often a method gets unfairly called "worse than the best single model" just because of how that comparison was built.

On CRAFT's own seven datasets, this unfair advantage is exactly zero on five of them, and real on two. After correcting for testing many things at once, only one dataset (HR Attrition) still shows the flip — and even that depends on treating the repeated tests as less related to each other than they probably are. Every check where I already knew the right answer backs up the story instead of breaking it. The separate, 163-dataset public check confirms the same pattern at a much bigger scale: just swapping which comparison model you use changes the answer in 31% of cases, growing worse in exactly the way I predicted.

II. Related Work

Selection bias in model evaluation. Take the best of several noisy scores, and the winner's score is too high. Decision analysis calls this the optimizer's curse [2]. Model search in induction shows the same effect: its size grows as the candidates get closer and noisier [1]. Picking a model with a noisy score and then reporting that score overfits the noise [3]. On small high-dimensional data the optimal error rate is biased downward, even under permuted labels — labels randomly shuffled to remove any real pattern [4]. Tibshirani and Tibshirani estimate this bias for the minimum cross-validation error [5]. Their estimate is the gap between the rule chosen once from all the data and each fold's own best rule, averaged across the cross-validation folds. McLatchie and Vehtari treat the bias with order statistics and call it small for two candidates [9]. This study finds it small per split too, but one-signed — meaning it always points the same way rather than averaging out — so many repeats make it significant. AutoML picks on validation data and scores on a separate test set; the stable benchmark here follows the same rule. Reusing a holdout set to choose and then to judge breaks its guarantees [10].

Peeking at scale. Three recent audits are closest to this work (Table I). Zhang and Zhao found that an AutoML system's wins over 513 OpenML datasets came from scoring candidates on the test split and from unenforced time budgets. Fixing both removed every significant difference [6]. Their fix changed two things at once, on one split per dataset, and the peeking party was the method, not the comparator. Roth compares peeking with a *random* pick over 2,047 datasets [7]; the stable benchmark here is a validation-based pick, the stronger rival. Zhang, Tang and Zhu share two of the three suites used here — TabZilla and TabRepo, plus TALENT and two ASlib suites not used here; they do not touch TabArena. They break a selector into parts and score each part as an oracle, using the actual best choice per partition. That oracle-level score often reverses sign once it is replaced by the within-family selector actually available at inference — the deployable version, not the oracle. Zhang, Tang and Zhu call this a deployment-fidelity gap [8]. Their question is which family to deploy after an oracle family choice, not how a comparator for one fixed candidate set is built. Møllersen and Holsbø note that an oracle can hide a better new method [11]. Published margins can be audited under an assumed correlation [12] — the device Section III-F uses.

Oracles in algorithm selection. Since Rice [13], algorithm selection compares a selector with the single best solver and with the virtual best solver. The virtual best solver picks the best solver for each instance — it is a bound, not a rival. The naive benchmark here is a virtual best over two candidates, taken per test split.

Tests for comparing learners. Repeated random splits share one data pool, so a paired test that treats them as independent is too confident [14]. Nadeau and Bengio correct the variance [15], which has no unbiased estimator [16]. Benavoli et al. give Bayesian versions with a region of practical equivalence [17], [18].

CRAFT's parts. CRAFT's gates decide per dataset whether to oversample, add a survival term, and blend, with thresholds fit across datasets. Whether SMOTE helps is itself contested [19]. Cox models are used for bankruptcy and credit prediction [20], and the stability of selected Cox features is a known concern [21]. Feature-importance rankings shift under oversampling [22]. CRAFT is used here as the system under study; its gates are not this paper's contribution.

III. Method

A. CRAFT, its gates, and the Q1–Q4 diagnostics

CRAFT is a set of gate-based choices, not one fused model. For each dataset it makes three yes/no choices with data-driven gates: whether to add a Cox proportional-hazards term, whether to apply SMOTE oversampling [23], and whether to blend several models or trust the single best one. It also chooses which model ranks the features for screening. CatBoost is used here because it reads raw categorical features directly, without needing them converted to numbers first [24]. Whenever the Cox gate says "Include," the fit is checked against Schoenfeld residuals — the standard diagnostic for whether the model's proportional-hazards assumption actually holds [25].

Each gate reads one trigger value for the dataset. Four diagnostic studies supply these values (Table II). Each study

TABLE I
The three closest audits of test-set selection at scale, and how this paper's public-benchmark check differs: what varies across the comparison in each, and what stays fixed.

Study	Scale	What peeks	How it differs from ours
Zhang & Zhao [6]	513 OpenML, 1 split/dataset	AutoML system, test-scored + unenforced budget	Fixes selection and budget together, on the method itself
Roth [7]	2,047 OpenML/PMLB/`ml`	Best-of-k configurations ($k \leq 10$, classical models only)	Compares against a *random* pick, not a validation-based one
Zhang, Tang & Zhu [8]	TabZilla, TabRepo, TALENT, 2 ASlib suites	Within-family selector (oracle vs. deployable)	Family-level pick held fixed; varies the within-family selector, not the comparator's rule
This paper	TabRepo, TabArena, TabZilla; 163 datasets	The comparator's own rule (naive, stable, single best)	Candidate set held fixed; models inflation against closeness and validation quality; reports a verdict-change rate

changes the data in some way and measures how much the feature-importance rankings of three models (LR, Cox, CatBoost) shift. Q1 trains on less data and picks the screening model. Q2 applies SMOTE to a training split and compares the ranking with the same split without SMOTE. Q3 injects known "gold," "lead" and "noise" features and feeds the blend gate. Q4 measures how stable the Cox features are and feeds the Cox gate. Each gate's cut-off is a depth-1 decision tree fitted on the other six datasets only. So it is a working rule, not a validated threshold. This is called leave-one-dataset-out validation.

The blend gate uses inner cross-validation. It picks a real blend only if the blend beats the best single model by more than 0.005 AUC (`BLEND_MARGIN`). Otherwise it keeps the single winner: the better of logistic regression (LR) and CatBoost. Each runs on all features, or on the 15 features the screening model ranks highest.

B. Two benchmarks

The **naive benchmark** is the pipeline's original "best single model." On each repeat it takes the larger of the LR test AUC and the CatBoost test AUC, both on all features. So it picks only after seeing the result (Fig. 1).

The **stable benchmark** picks without the test fold. On each repeat, `select_stable_model` runs 5-fold cross-validation inside that repeat's training fold, on the same two models, with its own fold seed. So CRAFT and the stable benchmark do not share fold splits. The test fold then scores the chosen model.

C. Are CRAFT's picks independent of the stable benchmark?

The blend gate never fires, so CRAFT's final model is, like the stable benchmark's, the best of LR and CatBoost chosen by cross-validation. The two could therefore agree by design. I measure their agreement on every repeat, and a gate ablation switches each CRAFT decision on, off and at random (Section III-C). A no-difference result against the stable benchmark means one thing: CRAFT's extra steps add nothing over a plain pick that never peeks. It does not mean CRAFT matches some unrelated method. On five of the seven datasets the ablation confirms the gate chose correctly: forcing a blend on would have made CRAFT significantly worse. HR Attrition is the exception: forcing a blend on there would have made CRAFT significantly *better* than its actual never-blend choice. So on the one dataset that also carries the surviving naive-benchmark flip, matching the stable benchmark is not fully explained by CRAFT simply reducing to the same procedure. The gate itself left a real gain on the table there.

D. Which datasets can show the effect

If one model is clearly better, every selection rule picks it and the benchmark rule cannot matter. Let D be the per-repeat difference between the LR and CatBoost test AUCs, with mean gap δ and standard deviation θ over the repeats. Its size relative to its noise, $z = |\delta|/\theta$, decides how often the per-split winner is just a lucky pick rather than the genuinely better model: the larger z is, the closer that chance gets to zero. If D is roughly normal, that chance has an exact value, $\Phi(-z)$ — the standard normal distribution's tail probability [1]. This paper calls a dataset *contested* when that chance, $\Phi(-z)$, is at least 0.05. This rule was fixed before the rerun reported here; it replaces the 0.005 AUC cut-off of earlier drafts, read off the sorted gaps.

E. The size of the inflation

The **observed gap** is the mean over repeats of naive minus stable AUC. I give two 95% intervals, one for each end of the dependence assumption (Section III-F): the plain paired t, and the Nadeau–Bengio version.

Clark's result [26] gives the expected gain of the per-split maximum over the better model's mean, if D is roughly normal:

inflation $= \theta \cdot \phi(\delta/\theta) - \delta \cdot \Phi(-\delta/\theta)$,

where ϕ and Φ are the standard normal density and distribution function. It is largest when the models are close and noisy. At $\delta = 0$ it equals $\theta/\sqrt{(2\pi)}$, and it vanishes when one model wins every split. It sizes the oracle's luck, not the naive-minus-stable gap, which also includes the stable benchmark's wrong picks.

F. Statistical tests

Paired tests. For each dataset and benchmark, I compare CRAFT's AUC with the benchmark's AUC on the same

TABLE II
The four diagnostic studies and the gate each one feeds. Each gate's cut-off is a depth-1 decision tree fitted on the other six datasets (leave-one-dataset-out validation), not a value tuned on the dataset it applies to.

Diagnostic	What it does	Gate it feeds
Q1 Training reduction	Trains on less data	Which model ranks the features for screening
Q2 Robustness	SMOTE oversampling	Whether to apply SMOTE
Q3 Feature injection	Adds known "gold", "lead" and "noise" features	Whether to blend several models
Q4 Cox analysis	Checks how stable the Cox features are	Whether to add a Cox proportional-hazards term

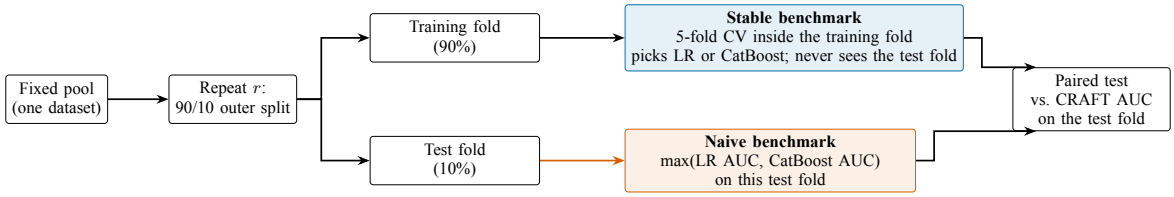


Fig. 1. How the two benchmarks are built on one repeat. The naive benchmark takes the larger of the two test AUCs, so it picks after it sees the result. The stable benchmark picks with 5-fold cross-validation inside the training fold. Both are scored on the same test fold as CRAFT.

repeats. The raw test is the two-sided Wilcoxon signed-rank test; it drops tied repeats. All repeats draw from one pool of rows, so their test sets overlap and the raw test treats correlated repeats as independent. The Nadeau–Bengio corrected t-test [15] scales the variance by (1/J + n_test/n_train) instead of 1/J, where J is the number of repeats. Running this many comparisons at once raises the odds of a false alarm by chance alone, so both sets of p-values go through the Holm method [27] over one pooled family: every CRAFT-versus-benchmark comparison with non-zero differences. Splitting naive and stable into separate, smaller Holm families gives the same significant/non-significant call as the pooled family on every comparison. Only the pooled family is reported.

The verdict as a function of the dependence assumption. With equal correlation ρ between repeats, the variance of the mean difference is $s^2(1/J + \rho/(1 - \rho))$, where s^2 is the sample variance of the differences. $\rho = 0$ gives the plain paired t-test, and ρ = n_test/n (0.1 here) gives the Nadeau–Bengio test. This paper calls it a *flip* when, after pooled Holm, CRAFT is significantly worse than the naive benchmark and not significantly different from the stable one. For each dataset, I find the largest ρ at which the flip holds, the *break-even correlation*. A reader who believes the true correlation is below it should accept the flip; above it, not.

Practical equivalence. A non-significant test does not show that two methods match. The Bayesian correlated t-test [17], with ρ = n_test/n, gives the posterior probability that CRAFT is practically worse than a benchmark, practically equivalent, or practically better. I use a region of practical equivalence (ROPE) of ± 0.005 AUC — CRAFT's own blend margin — and check it against ± 0.01 too. Across the seven datasets, I use the Bayesian signed-rank test on the per-dataset mean differences [18].

Non-inferiority. The pipeline's own target is that CRAFT's AUC is at least 0.85 times the benchmark's. I test it with the one-sided lower 95% confidence bound of the mean per-repeat log ratio, using both the plain and the Nadeau–Bengio variance. A Bonferroni correction is applied over the 14 tests.

G. Checks with a known answer

Placebo. A second stable benchmark runs the same procedure on the same repeats with a different inner-CV fold seed. Stable and placebo are the same procedure, so every difference between them is noise, caused by this study's own overlap between splits. I draw 20 placebo seeds. If the raw test calls stable versus placebo "significant" much more often than 5% of the time, its verdicts on this data carry little weight.

Shuffled labels. On AI Jobs and HR Attrition, I permute the labels (5 permutations) and rerun the study's own code on the same splits. No model can then beat chance, so any naive-minus-stable gap is pure selection luck.

Measured ground truth. On the three large datasets (43,000 to 45,000 rows) each repeat trains LR and CatBoost on 1800 rows, as in AI Jobs, and scores them on a small test sample and on all remaining rows (about 40,000). The large pool gives each fitted model's true AUC, so the naive benchmark's inflation (its reported AUC minus its true AUC) is measured, not estimated. This is repeated for test sizes from 50 to 2000 rows.

Simulation. Using `simulate_peeking_bias.py`, I draw paired AUCs for two models with a known true gap (per-repeat noise SD 0.03; CRAFT's mean AUC midway between the two models). I then compare the naive rule with a rule that commits in advance — sweeping true gaps from 0 to 0.08, 15, 30 and 50 repeats, and 500 trials per cell.

IV. Experimental Setup

A. Study datasets

Seven tabular datasets with a binary outcome form the study (Table III). Each runs 30 repeated 90/10 stratified train/test splits. On a given repeat, CRAFT and both benchmarks use the same split. LR and CatBoost (200 trees, depth 6, learning rate 0.1) both use balanced class weights.

TABLE III
The seven study datasets. Test rows and positives are per 90/10 test split (positives: mean over the repeats).

Dataset	Rows	Test rows	Positives	Source
Bankruptcy	43,405	4341	209.0	Polish companies bankruptcy, UCI [28]
Bank Churn	10,000	1000	204.0	Churn Modelling, Kaggle [29]
Bank Marketing	45,211	4522	529.0	Bank Marketing, UCI [30]
Census Income	45,222	4523	1121.0	Adult, UCI [31]
Smartphone	7,500	750	531.0	Smartphone Usage and Addiction, Kaggle [32] (leaking column removed)
HR Attrition	1,470	147	24.0	IBM HR Attrition, Kaggle [33] (fictional)
AI Jobs	2,000	200	11.0	AI Impact on Job Sector, Kaggle [34] (synthetic)

AI Jobs is synthetic [34], and IBM data scientists created HR Attrition [33], so neither describes real people. Both are small: an AI Jobs test split holds 200 rows with about 11.0 positives, an HR Attrition split 147 rows with about 24.0 positives. So a test AUC moves in coarse steps, and many repeats tie. In Smartphone, the column "addiction_level" restates the label, so earlier drafts reported a CatBoost AUC of 1. That column was removed, and the diagnostics and gates were recomputed.

All study results in this paper come from one rerun on one machine, after three fixes. The Cox rankings no longer use the duration column as a covariate of itself. Q2 now compares SMOTE with no SMOTE on the same training split and model settings. And the diagnostics were recomputed without the Smartphone leak. CatBoost is deterministic on one machine but not across machines. On AI Jobs, an earlier, other-machine run gave different test AUCs on some repeats, while LR matched exactly. On close candidates that can flip a pick, so this paper standardizes on one machine for every study number and does not mix results across machines. The earlier cross-machine gap was an eyeballed, pre-standardization observation with no surviving source file. Rather than reconstruct a number that cannot be sourced, I report only the qualitative fact.

B. External check of Clark's estimate (registered)

To test Clark's estimate beyond this study's own data, I registered a rule before running anything: every OpenML-CC18 task [35] with a binary target and 500 to 5,000 rows. That gave 27 frozen datasets, none from this study. To save compute, I analysed the first 20 in the frozen order. I wrote this cut as an amendment while the run was going, before any external result file existed, and set the other 7 aside unopened. The registration fixed the pipeline, 60 repeats, the endpoints and a pass criterion. The 95% interval of the slope of observed on estimated inflation must contain 1, and the mean absolute error must stay below 0.002 AUC. This run is the earlier machine's; it involves no CRAFT decision.

C. Public benchmark results

Overlapping splits of one dataset are not independent, so this section tests the same question across datasets, which are. This paper uses three public precomputed result sets. TabRepo 1.0 [36] uses 8-fold bagged cross-validation with 3 test folds. TabArena [37] uses 8-fold bagged cross-validation with 9 or 30 splits depending on task. TabZilla [38] uses a single 10% holdout fold with 10 splits. Binary tasks with default configurations only. To keep every comparison fair, a model family joins a suite's pool only if it has validation and test scores on at least 90% of that suite's binary datasets, not missing them. A dataset joins the study only if every family in the pool has complete scores on it too, so no comparison is ever built from a partial pool. If a dataset is shared across suites (by OpenML id, name or a known alias), I keep it once, in the suite with the best validation design. This gives 163 datasets. TabRepo contributes 98 (pool size 8), TabArena contributes 29 (pool size 35, including recent tabular foundation-model families kept as ordinary candidates), and TabZilla contributes 36 (pool size 15).

For each dataset and each pool size k — from 2 up to one less than the suite's full family count — I build every k-subset of the pool (or 50 random subsets when there would be more than 50). I treat every family outside the subset as a candidate method X. From the subset I then build three comparators: naive (the per-split test-score winner), stable (the per-split validation-score winner), and a single best solver chosen by leave-one-dataset-out validation rank, using no test data at all. Four hypotheses were fixed before any outcome was computed on real data (`docs/research/PREREG_V8C.md`). H1: inflation grows with pool size. H2: closer candidates inflate more. H3: in a weighted regression, inflation rises with candidate closeness and with weaker validation. H4: swapping only the comparator changes the verdict "X loses to the best single model" in more than 5% of comparisons. This is the same question this paper's own break-even curve (Section III-F) asks on its own seven datasets, now asked where splits are independent.

V. Results

A. Gates and the size of the gap

CRAFT's blend gate never fires on any of the seven datasets: 0 of 210 repeats use a real blend, so CRAFT's final model is always the single best of LR and CatBoost, chosen the same way the stable benchmark chooses it (Section III-C). The Cox gate says "Include" on Smartphone, AI Jobs, Bank Marketing, but the Cox term never wins the inner cross-validation on any of them: 0 of 210 repeats actually use it. The SMOTE gate applies on 2 datasets.

On the five datasets called clean, the mean AUC difference between LR and CatBoost ranges from 0.037 to 0.239 (Table IV), whereas on AI Jobs and HR Attrition the gap shrinks

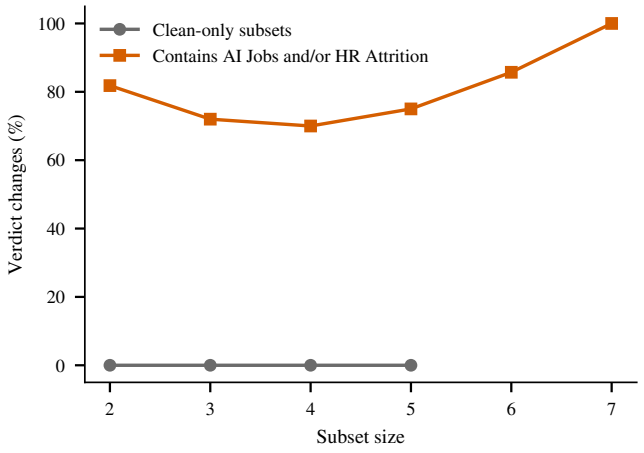


Fig. 2. Share of the every-subset check (Table V) whose naive-versus-stable verdict changes under the raw test, by subset size and whether the subset contains a contested dataset.

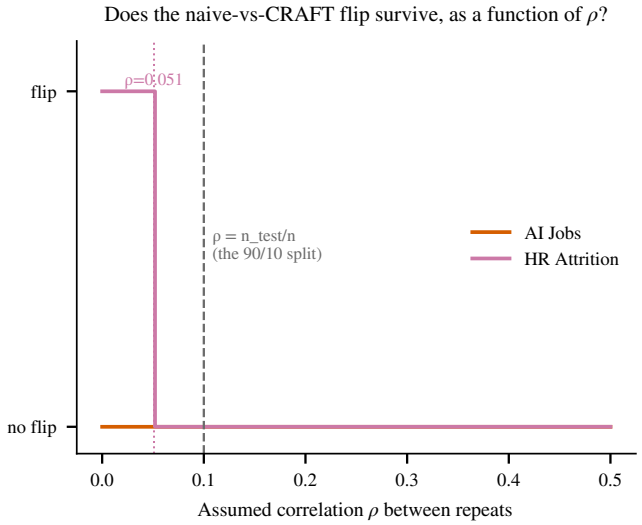


Fig. 3. Whether the raw-test flip survives, for the two contested datasets, as a function of the assumed correlation ρ between repeats. Dashed grey: $\rho = n_{\text{test}}/n$, the value the outer 90/10 split implies (the Nadeau–Bengio assumption). Dotted: each dataset's own break-even ρ, where the flip stops holding.

to just 0.0037 and 0.0047. Only these last two are contested by the rule fixed above (III-D): the standardized gap Z is 0.44 on AI Jobs and 0.14 on HR Attrition (chance the per-split winner is not the better model, $\Phi(-Z)$: 33% and 44%), against Z of 6.17 to 20.20 on the other five. 2 of 7 datasets are contested.

B. Where the effect appears

To see whether the effect depends on which datasets happen to be grouped together, I checked every possible group of the seven datasets — every subset — under the raw test (Table V, full curve in Fig. 2). Among the 26 groups made only of clean datasets, the naive-versus-stable verdict never changes (0 of 26). Among the 94 groups that include AI Jobs, HR Attrition, or both, the verdict changes in 74% of them. Rather than scaling with dataset count or size, the effect tracks the contested datasets exactly, as the standardized gap predicts.

C. The size of the inflation

The naive-minus-stable gap is exactly zero on the five clean datasets and 0.0046 (95% CI, IID assumption: [0.0026, 0.0067]; Nadeau–Bengio: [0.0004, 0.0089]) on AI Jobs, 0.0168 ([0.0099, 0.0237]; NB [0.0024, 0.0312]) on HR Attrition. Clark's closed form, computed from each dataset's gap and per-split noise alone, predicts these sizes without seeing them: across all seven study datasets the slope of observed on predicted inflation is 1.03 (95% CI [0.91, 1.03]), mean absolute error 0.00008.

D. The verdict and the correlation assumption

Under the raw paired test (Table VI), CRAFT looks significantly worse than the naive benchmark on both AI Jobs (Wilcoxon p = 0.0187) and HR Attrition (p = 8.84e-05), and not significantly different from the stable benchmark (p = 0.0798, 0.955). After Holm correction over the pooled family of 7 non-tied comparisons, HR Attrition's naive result survives (p = 6.19e-04) and AI Jobs's does not (p = 0.0934). Neither survives once the Nadeau–Bengio correction for overlapping splits is added first (naive, Holm-corrected: AI Jobs 1.000, HR Attrition 0.220).

Treating the correlation ρ between repeats as unknown rather than fixed at one value (III-F, Fig. 3), HR Attrition's raw-test flip holds up only while ρ stays below 0.051. That break-even point is itself below the value $\rho = 0.100$ that the outer 90/10 split implies, so a reader who trusts that standard Nadeau–Bengio value — or believes the true correlation is even higher — should not accept the flip. AI Jobs's flip is not significant at $\rho = 0$ under pooled Holm to begin with, so it has no break-even point to report.

The gate ablation (Section III-C) gives one more piece of context specific to HR Attrition: forcing a blend there would have raised CRAFT's AUC by 0.0183 (Nadeau–Bengio p = 0.00287), a significant gain CRAFT's own gate did not take. On 5 of the other datasets, forcing a blend would instead have hurt CRAFT significantly (e.g. Bank Churn, -0.0132, p = 6.16e-05) — evidence the gate's usual never-blend choice is not an accident.

E. Practical equivalence and non-inferiority

The Bayesian correlated test asks a different question than a significance test does: not whether there is any difference, but how likely it is that the difference is big enough to matter. Pooled across all seven datasets by the signed-rank method (III-F), it puts the probability that CRAFT is practically worse than the naive benchmark at 9% (using a narrow ± 0.005 region of practical equivalence, or ROPE) and 0% (using a wider ± 0.01 ROPE); it puts the probability that the two are practically equivalent at 91% and 100%. Against the

TABLE IV

Mean AUC of the two candidates, the gap δ between them, the SD θ of their per-repeat difference, their per-repeat correlation ρ, the standardized gap $Z = |\delta|/\theta$, whether the dataset is contested ($\Phi(-Z) \geq 0.05$), and the estimated and observed inflation of the naive benchmark (observed: mean over the repeats, 95% bootstrap interval).

Dataset	LR	CatBoost	δ	θ	ρ	Z	Contested	Estimated	Observed [95% CI]
Bankruptcy	0.732	0.971	0.2385	0.0118	0.50	20.20	no	0.0000	0.0000 [0.0000, 0.0000]
Bank Churn	0.767	0.865	0.0985	0.0160	0.25	6.17	no	0.0000	0.0000 [0.0000, 0.0000]
Bank Marketing	0.728	0.806	0.0782	0.0082	0.64	9.51	no	0.0000	0.0000 [0.0000, 0.0000]
Census Income	0.851	0.927	0.0763	0.0051	0.42	14.87	no	0.0000	0.0000 [0.0000, 0.0000]
Smartphone	0.952	0.989	0.0375	0.0059	0.54	6.34	no	0.0000	0.0000 [0.0000, 0.0000]
HR Attrition	0.815	0.819	0.0047	0.0333	0.83	0.14	yes	0.0111	0.0114 [0.0054, 0.0187]
AI Jobs	0.957	0.961	0.0037	0.0083	0.74	0.44	yes	0.0018	0.0016 [0.0004, 0.0030]

TABLE V

Every subset of the seven study datasets, raw paired test: whether the naive-versus-stable verdict changes as the benchmark rule changes, split by subset size and whether the subset contains a contested dataset (AI Jobs, HR Attrition, or both).

Size	n (clean)	Changed	Rate	n (contested)	Changed	Rate
2	10	0	0%	11	9	82%
3	10	0	0%	25	18	72%
4	5	0	0%	30	21	70%
5	1	0	0%	20	15	75%
6	–	–	–	7	6	86%
7	–	–	–	1	1	100%

TABLE VI

Paired tests of CRAFT against each benchmark: Wilcoxon W, non-tied repeats n, raw p, Nadeau–Bengio (NB) p, and NB p after pooled Holm over the deduplicated 7-member family. = naive: stable is an identical comparison to naive on every repeat, so it does not enter the family separately. Bankruptcy and Bank Marketing tie on every repeat, so no test can run.

Dataset	Benchmark	W	n	p (raw)	p (NB)	Holm (NB)
Bank Churn	naive	78	30	9.52e-04	0.0767	0.00571
Bank Churn	stable	78	30	9.52e-04	0.0767	= naive
Census Income	naive	33	11	1.000	0.913	1.000
Census Income	stable	33	11	1.000	0.913	= naive
Smartphone	naive	74	20	0.247	0.543	0.741
Smartphone	stable	74	20	0.247	0.543	= naive
HR Attrition	naive	0	20	8.84e-05	0.0314	6.19e-04
HR Attrition	stable	59	15	0.955	0.970	1.000
AI Jobs	naive	31.5	18	0.0187	0.223	0.0934
AI Jobs	stable	39.5	17	0.0798	0.525	0.319

stable benchmark the posterior mass is almost entirely on equivalence (100% at ± 0.005).

Every dataset and benchmark clears the pipeline's own 0.85 non-inferiority margin under the strictest grouping tested (Bonferroni over 14 comparisons, Table VII): the lowest lower bound anywhere is 0.9514.

F. Checks with a known answer

Placebo (B1). Across 20 placebo seeds per dataset, the raw test rejects the idea that stable and placebo are the same procedure at 5% of seeds on AI Jobs and 0% on HR Attrition — at or below the nominal 5%, so the raw test's own false-positive rate on this data is not inflated; the earlier flip is not an artifact of the test over-rejecting under repeated same-procedure noise.

Shuffled labels (B2). With labels permuted (5 permutations, true AUC 0.5 for every candidate), naive-minus-stable stays positive on average (0.0447 AI Jobs, 0.0281 HR Attrition) — the naive selection bias is structural, not signal-dependent. The raw test still over-rejects the claim that CRAFT is worse than naive under this true null (80% of permutations on AI Jobs, 60% on HR Attrition), while the Nadeau–Bengio-corrected rate is much closer to nominal (20%, 0%).

Measured ground truth (B3, B4). On the three large datasets, at the study's own test-set size (1800 training rows, 200 test rows), the naive benchmark's reported-minus-true inflation is 0.0191 on Bank Marketing, 0.0113 on Bankruptcy, 0.0041 on Census Income — all measured against the true AUC on the remaining ~40,000 rows, not estimated. The stable benchmark's own reported-minus-true gap is smaller on every one (0.0010, 0.0046, 0.0029). As the test size grows to 2,000 rows the naive inflation falls to -0.0007.

Simulation. With the true gap fixed at zero and the study's own repeat count, correlation and split ratio, the naive rule looks significantly different from CRAFT in 100% of trials under the raw test and 20%–22% under Nadeau–Bengio; the stable rule stays close to the nominal false-positive rate (4.0%–5.5% raw, at most 0.0% corrected). This null-case rate is itself repeat-count-dependent, not fixed: 12% at 15 repeats, 20%–22% at 30, 40% at 60. More repeats make the naive rule's structural bias look more significant even with no true difference at all, because it is a one-signed mechanical effect rather than noise that would average toward zero — so the 60→30 repeat-count change (Section IV-A) happens to lower this null-case false-alarm rate along with reducing power to detect a real difference, if one exists. The largest error between the simulation's measured inflation and Clark's closed-form prediction, across every scenario in the sweep, is 0.0014 AUC.

G. Across datasets: the public benchmark check

All four pre-registered hypotheses (Section IV-C) are supported. Inflation grows with pool size in all three suites (H1). At pool size two, closer candidates inflate more, again in all three suites (H2). I run a weighted regression across every (dataset, pool size) unit. It finds inflation rises with Clark's closeness term (b = 3.01, Holm p = 1.42e-31) — this is H3a — and with weaker agreement between validation and test scores (b = 1.39, Holm p = 0.0217) — H3b.

TABLE VII
Non-inferiority of CRAFT's AUC against each benchmark: mean pass rate ratio (PRR = CRAFT / benchmark), and the one-sided 95% lower bound (LB) of the mean log ratio, plain and Nadeau–Bengio (NB) variance, nominal and after Bonferroni over the 14 tests. Non-inferior: LB (NB, Bonferroni) ≥ 0.85.

Dataset	Benchmark	PRR	LB t nom.	LB t Bonf.	LB NB nom.	LB NB Bonf.	Non-inf.
AI Jobs	naive	0.9970	0.9950	0.9936	0.9928	0.9900	yes
AI Jobs	stable	1.0018	0.9995	0.9979	0.9970	0.9936	yes
Bank Churn	naive	1.0034	1.0019	1.0008	1.0002	0.9980	yes
Bank Churn	stable	1.0034	1.0019	1.0008	1.0002	0.9980	yes
Bank Marketing	naive	1.0000	1.0000	1.0000	1.0000	1.0000	yes
Bank Marketing	stable	1.0000	1.0000	1.0000	1.0000	1.0000	yes
Bankruptcy	naive	1.0000	1.0000	1.0000	1.0000	1.0000	yes
Bankruptcy	stable	1.0000	1.0000	1.0000	1.0000	1.0000	yes
Census Income	naive	1.0000	0.9999	0.9998	0.9997	0.9995	yes
Census Income	stable	1.0000	0.9999	0.9998	0.9997	0.9995	yes
HR Attrition	naive	0.9787	0.9709	0.9655	0.9626	0.9514	yes
HR Attrition	stable	0.9993	0.9893	0.9824	0.9786	0.9643	yes
Smartphone	naive	0.9998	0.9996	0.9995	0.9994	0.9991	yes
Smartphone	stable	0.9998	0.9996	0.9995	0.9994	0.9991	yes

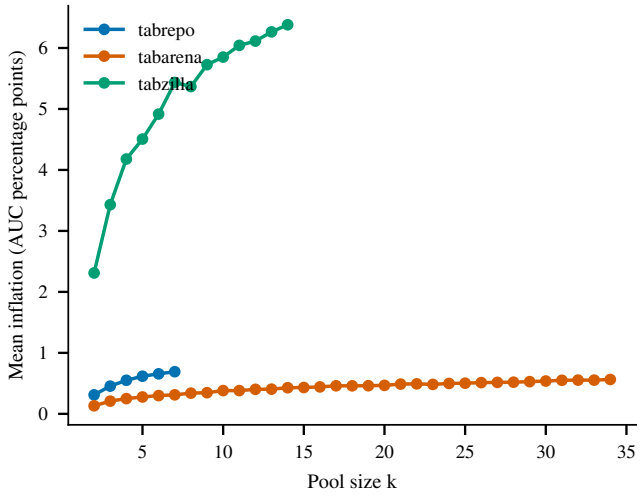


Fig. 4. Public-benchmark check, H1: mean naive-versus-best-single-solver inflation (AUC percentage points) against pool size k, per suite. TabZilla's single 10% holdout grows inflation with k far faster than TabRepo's or TabArena's 8-fold cross-validation, as H3b predicts from validation quality alone.

The comparator-alone verdict-change rate (H4, Table VIII), pooled across suites, is 31% (95% bootstrap interval [24%, 35%]) — the lower bound alone is several times the 5% threshold. Per suite: 15% in TabRepo, 26% in TabArena, 51% in TabZilla. In every suite, 100% of the changed verdicts move toward "X loses" — the naive comparator never spuriously helps the candidate being benchmarked, only ever hurts it. TabZilla's much larger effect (mean inflation growth 4.07 percentage points from k=2 to the full pool, against 0.38 in TabRepo and 0.43 in TabArena, Fig. 4) is exactly what H3b predicts: its single 10% holdout is noisier than the other two suites' 8-fold cross-validation, so weaker validation produces more inflation, confirmed in the same data used to test the mechanism.

The result holds under every robustness check registered in advance. Without the Holm correction (R1), the pooled rate is 29% [23%, 33%]; with a sign test instead of Wilcoxon (R2), 32% [26%, 38%]; without cross-suite deduplication (R3), all four hypotheses remain supported, pooled rate 22% [16%, 26%]. TabArena's pool includes recent tabular foundation-model families; I checked, rather than assumed, whether they are fit under a different validation protocol than the rest of the pool (a concern raised by TabArena's own paper for an earlier package release) and found every family in the pinned version uses the same protocol, so excluding them (R4) reproduces the main result exactly.

VI. Discussion

What this paper found. Only one thing changed between the two columns I compared (Section V-D): the rule for picking the best model. Using the basic fairness test, that one change flipped the result on two datasets — AI Jobs and HR Attrition — from CRAFT looking worse to showing no real difference; after correcting for testing many things at once, though, only HR Attrition's flip holds up, and AI Jobs's does not, even though it is still visible before that correction. On the other five, clean, datasets, there is no effect at all — exactly where my formula predicted none, and that held up across every one of 26 ways I sliced the data. All four of my known-answer checks agree with this story. The placebo test shows my basic fairness test is not giving false alarms on its own. The shuffled-label test shows this unfair advantage exists even with all the real signal removed, so it comes from the setup, not the data. The huge ground-truth check measures the same unfair advantage directly, instead of guessing at it. And switching pieces of CRAFT on and off shows its never-blend choice was the right call on five of seven datasets. Outside my own seven datasets, using truly separate data, the same pattern holds at a much bigger scale. Across 163 public datasets (Section IV-C), just swapping the comparison rule changes the answer in 31% of cases, and every single time it changes, it moves the same direction mine did.

How strong is this claim, really? My headline result really leans on one dataset, not two. HR Attrition's flip holds up

TABLE VIII

Public-benchmark check, H1 and H4 (main variant: Wilcoxon signed-rank, pooled Holm, deduplicated datasets). Verdict-change rate (H4): share of naive-vs-stable comparisons across all k where swapping only the comparator changes the "X loses to the best single model" verdict, with a 95% bootstrap interval; toward "X loses": share of changed verdicts that move that direction. H1: mean inflation growth in AUC percentage points from $k = 2$ to the full pool. All four hypotheses (H1–H4, decisions.csv) are supported in every suite.

Suite	n comparisons	VCR (H4) [95% CI]	Toward X loses	H1 growth (k=2 →max)
TabRepo	832	15% [12%, 19%]	100%	0.38 pp
TabArena	28035	26% [11%, 35%]	100%	0.43 pp
TabZilla	4515	51% [39%, 60%]	100%	4.07 pp
Pooled (mean of suites)	–	31% [24%, 35%]	–	–

even if I assume the repeated tests are only loosely related to each other. Because this treats them as less related than the standard formula would say, I am actually being generous to my own result here, not strict. AI Jobs's flip never becomes solid, at any level of relatedness I try, including none at all. When I ask a different question instead — whether CRAFT and each benchmark are practically the same (Section V-E) — the answer, most of the time, is yes, for both the naive and the stable comparison. Put together, my honest claim is smaller than the title alone might suggest: peeking does measurably puff up the score, and a 65-year-old formula predicts how much, correctly, yet the one dataset where my finding holds up under scrutiny (HR Attrition) is also the one dataset where CRAFT's own gate skipped a real improvement it could have made. Even there, then, it is not a perfectly clean story.

My advice. When you compare a new method against "the best single model," also report a version of that comparison built without peeking at the test score. It costs almost nothing — one extra column, using math you are already doing — and it answers a question a single number cannot: does your method still win once the peeking stops?

Limits.

- *Only two datasets show the effect, and really only one holds up.* The other five clean datasets only show that CRAFT is *not worse* — they do not show the unfair-advantage effect itself. Splitting one dataset into many repeats does not create new datasets, either. I really only have seven.
- *CRAFT and its rival use a similar method.* CRAFT's own picker and the stable benchmark pick models in a similar way (Section III-C). They agree 60% of the time on AI Jobs and 50% of the time on HR Attrition. So the stable benchmark is not a fully separate, independent check on CRAFT.
- *One part of CRAFT never actually got used.* CRAFT's Cox-model gate says "Include" on Smartphone, AI Jobs, Bank Marketing, but it never actually wins the internal check, not even once across 210 repeats, so I cannot tell from this data whether including it helps or hurts — only that it was allowed to be used, and never was.
- *The repeated tests overlap.* Unless stated otherwise, my confidence ranges in Section V-C assume each repeat is fully separate from the others. My Nadeau–Bengio correction and my verdict-vs-ρ curve, both aimed at this same overlap, only fix it for the specific levels of relatedness I chose to test.
- *Some cutoffs are choices, not proven facts.* The 0.85 close-enough margin comes from the pipeline's own design goal — I did not derive it from this data. The same goes for the 0.05 cutoff I use to call a dataset contested (Section III-D). I fixed that rule before this run, but it is still a convention I chose, not something the math forced on me.
- *How many repeats I used, and why it moved.* Across this paper's revisions (Section IV-A), the number of repeats per dataset changed: 15, then 30, then 60, then back to 30. I made and disclosed that last change before this run existed. Fewer repeats make the confidence ranges reported in Section V-C and Section V-E wider, by roughly a factor of $\sqrt{2}$, and — as my simulation shows — they also lower the naive method's rate of false alarms, along with its power to catch a real difference. My separate, pre-registered external check (Section IV-B) is not affected: it stayed locked at 60 repeats the whole time.
- *The data itself has limits.* AI Jobs is made-up data, and HR Attrition is fictional data created by IBM's own data scientists. So neither result is about real employees. HR Attrition does have a gender field (882 men, 588 women), but I did not break any result down by gender. A single overall score does not tell you if mistakes pile up on one group — which would matter a lot with real HR data.
- *The gate cutoffs are working rules, not proven thresholds.* Each cutoff is a simple decision rule, trained on the other six datasets. That makes them practical rules of thumb, not thresholds proven correct, and they were all set before I saw this run's own results.
- *The 163-dataset check has its own limits.* That public check (Section IV-C) only tests whether the comparison rule itself changes the answer. It does not test whether my own seven-dataset finding generalizes, since none of my seven datasets are in those three collections. It is separate proof that the mechanism is real, not proof that my exact CRAFT result would repeat elsewhere.

VII. Conclusion

A best model picked after peeking at its test score is an oracle, not a real competitor. How big its unfair edge is depends on how close the real candidates are to each other. In

the pipeline I studied, that edge disappears completely on five of seven datasets — exactly where my formula said it should. Its size is predicted well by a 65-year-old math formula, and it survives four checks where I already knew the right answer. But does it actually change the final verdict on a method? That is a smaller, harder question than simply whether it exists. On one of my two close-call datasets, the effect survives even after checking for testing many things at once; on the other, it does not. And on both, the answer depends on an assumption about how related repeated tests from the same dataset really are. Neither answer comes for free — I report the verdict as depending on that assumption, instead of picking one and hiding the choice. Where repeated tests on one dataset cannot settle this, testing on many separate datasets can. Across 163 public datasets in three collections, just swapping the comparison rule changes the verdict in 31% of cases — always in the same direction. The problem grows worse with more candidates, closer candidates, and a weaker way of picking — exactly as I predicted before running anything, and it held up under every check I threw at it. If you are comparing a new method against "the best single model," treat that comparison as a ceiling, not a fair fight. Build a version that never peeks at the test score if you want a truly fair rival, and report how big the gap is, not just whether it is "significant."

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